

Information Isomorphism Epistemology: From the Factor Hierarchy Law to the Reconstruction of Knowability

—How the Establishment of an Original Epistemological Operating System Renders Kant's "Thing-in-Itself" Unnecessary

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Abstract

This paper presents the establishment process of an original epistemological system—Information Isomorphism Epistemology—and its philosophical consequences for Kantian agnosticism. This system consists of four components, each with an independent origin and irreplaceable demonstrative function. The Law of Information Isomorphism reveals that in complex systems based on objective data and exhibiting regime switches, statistical structures carry information about regime structures—the Chow F achieves a globally unique maximum at the precise phase transition temperature, with deviation strictly zero in the analytic sense. The Factor Hierarchy Law reveals that any complex system is organized by a hierarchical structure of regime factors and execution factors—this is a cross-disciplinary, universal, formal meta-law of meta-laws that emerged from financial data and has been repeatedly verified across six completely independent disciplines. The Principle of Testability stipulates that any claim about factor importance in multi-factor systems must pass complete regime-dependence tests—claiming that "this factor is important" without specifying "under which regime" is an incomplete proposition. The Norms of Testability provide standardized testing procedures across six functional dimensions—functional dimensions remain relatively stable while implementation tools are replaceable. The four components, each independently originated, together constitute a complete epistemological system. The establishment of this system was not aimed at refuting agnosticism. Its starting point was a concrete methodological question. However, after its completion, a previously unforeseen philosophical consequence emerged: within the domain of scientific practice, the Law of Information Isomorphism replaces the function of the "thing-in-itself" as a specific explanatory item; the Principle of Testability blocks the retreat of "this is merely your subjective illusion"; the Norms of Testability provide a complete set of testing procedures and negative controls, making scientific practice equipped with this operating system no longer need to appeal to an unknowable "thing-in-itself" to guarantee its objectivity. The core argument of this paper can be summarized as "tool replacement"—not proving Kant wrong through philosophical debate, but using an original epistemological operating system to render the "thing-in-itself" no longer bearing any indispensable function in scientific practice.

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1. Introduction: The Birth of an Epistemological System Independent of Kant

1.1 Starting Point: A Concrete Methodological Question

The starting point of this system was not the grand philosophical question of "whether the world is knowable." Its starting point was a concrete, technical methodological question: in multi-factor complex systems, how can one reliably assess the importance of factors?

In traditional multi-factor research—from the Fama-French three-factor model to various asset pricing studies—researchers manually select factors, place them into the same linear equation, and compare whose coefficients are more significant. This research paradigm has a default assumption that has never been questioned: all factors are parallel. Their functions are identical—all directly explain the dependent variable. This default has never been argued for; it is simply silently assumed in every regression.

Tang (2026e) systematically tested this default for the first time. Across fifteen global equity markets, all candidate factors were subjected to exhaustive testing. The results revealed a phenomenon never previously noticed: the direct incremental R^2 of the China-US interest rate spread was close to zero, but the interaction incremental R^2 was 1.78 times the direct incremental R^2 . This means that "the function of some factors is not to directly explain the dependent variable, but to regulate how other factors explain the dependent variable." This was not derived from philosophical speculation—it was revealed by the data itself.

1.2 The Emergence of the Factor Hierarchy Law

This discovery gave birth to the Factor Hierarchy Law (Tang, 2026g, 2026r): in any complex system driven by multiple quantifiable factors, factors are naturally divided into two levels—regime factors and execution factors. Regime factors determine which set of rules currently applies, and their function is not reflected in direct explanatory power but in the interaction effects through which they regulate the exposure coefficients of execution factors. When a regime factor crosses a critical value, the factor weights of the entire system are systematically reset.

The Factor Hierarchy Law is not a concrete empirical regularity (such as $F=ma$ or the periodic table). It is a meta-law of meta-laws about how laws are organized—it subsumes derived meta-laws such as Noether's theorem, the principle of least action, and Shannon's information theory, providing them with a unified structural template. At the same time, it is itself also a meta-law: it reveals that "any complex system is organized by a hierarchical structure of regime factors and execution factors."

The Factor Hierarchy Law was not an accidental discovery emerging in a single discipline. It was subsequently repeatedly verified across five completely independent disciplines, and in each discipline, it actively discovered new structures that traditional theoretical frameworks could not accommodate. These discoveries fall into two categories. The first category consists of post-hoc verifications on truth-known systems—detecting precise statistical signals at known physical structures, proving that information isomorphism holds precisely, such as locking onto the positions of known particles with professional phenomenological fitting precision in particle physics. The second category consists of ex-ante discoveries on truth-unknown systems—actively discovering previously unknown real structures without any physical knowledge, which constitutes the most direct strike against agnosticism, such as the Tang Break, the 3D Ising double-peak fission, and chemical predictions.

In statistical physics, the precise information-theoretic duality between the Factor Hierarchy Law and the Ehrenfest classification of phase transitions received analytic mathematical proof on the 2D Ising exact solution, with deviation strictly zero in the analytic sense (Tang, 2026k). On the 3D Ising model—a problem strictly proven to have no analytic solution and to be NP-complete—the algorithm, without any physical knowledge, automatically discovered a hidden structure never previously revealed by traditional finite-size scaling analysis: the Bai-Perron double-breakpoint blind test automatically locked $T_1=4.39$ and $T_2=4.49$, precisely aligning with the switch of spontaneous magnetization M_{sp} and the switch of the Binder cumulant U_4 , with deviations of 0.01 and 0.01 respectively (Tang, 2026l). This discovery was later independently replicated across three lattice sizes on the 2D Potts $q=4$ model (Tang, 2026m).

In astronomy, stellar spectral type as a regime factor triggers a regime switch in the transit depth equation at a temperature of 4762K—the Tang Break. Eight tools—Chow test, Bai-Perron, CUSUM, MOSUM, PELT, Bayesian change point detection, HMM, and kernel change point detection—converged within the same temperature interval, with five independent physical dimensions all exhibiting structural ruptures (Tang, 2026h–j). This discovery directly challenged the standard planet radius assumption that had been used in astronomy for decades—transit depth is not an intrinsic property of the planet, but a relational quantity coupled by the planet, stellar radiation field, and detector response, with the instrument being merely a proxy regime factor for the stellar radiation field.

In chemistry, the Calcium-Lanthanum Singularity Theorem revealed a quantitative correspondence between regime switch intensity and orbital angular momentum— $F(l) \propto l^2$, with the f-block/d-block intensity ratio of 2.20 deviating only 2.2% from $(3/2)^2=2.25$ (Tang, 2026s). The operating system provided purely deductive numerical predictions for the yet-unsynthesized elements 119 and 120—element 119 density 3.3–3.5 g/cm³, element 120 density 7.0–7.5 g/cm³—with all parameters locked from the preceding 118 elements, none adjusted artificially for the eighth period (Tang, 2026t). The Chemical Gauge Group Theorem rigorously

derived the necessity of s, p, d, f orbitals from eight mutually exclusive regime states via SO(3) irreducible representations (Tang, 2026u).

In particle physics, color charge was confirmed as a meta-regime factor. The algorithm, without any knowledge of the physical meaning of charm and bottom quarks, automatically locked onto the positions of charmed baryons and bottomonium with professional phenomenological fitting precision—the charm-region breakpoint at 2.2865 GeV deviated only 0.00004 GeV from the charmed baryon PDG standard value of 2.28646 GeV, and the bottom-region breakpoint at 9.4603 GeV deviated only 0.0001 GeV from the bottomonium PDG standard value of 9.46030 GeV (Tang, 2026n). The strange region was confirmed as a third regime center, and the precursor cascade was precisely quantitatively verified—strange $T_2 \approx$ charm T_1 , charm $T_2 =$ bottom precursor (Tang, 2026o). The Gauge Group Structure Theorem derived the inevitable emergence of $SU(3) \times SU(2) \times U(1)$ from regime boundaries (Tang, 2026p–q).

In nuclear physics, N and Z were confirmed as semi-continuous regime factors ($\eta > 10^6$). The confirmed breakpoint network expanded from 5 to 7, with traditional magic numbers systematically shifted—N=20 and N=50 were not selected into the optimal model (Tang, 2026v). N=14 was cross-validated by three independent tools as a statistically significant structural rupture—a position never previously marked by traditional nuclear physics (Tang, 2026v). The shell model coverage was merely 15%—only 3 out of 20 blind breakpoints fell within the ± 2 range of traditional magic numbers (Tang, 2026w). The triple demotion demonstrated that the shell model, the liquid drop model, and the independent-particle-motion axiom are all projections of the quantum-classical hybrid under specific limits (Tang, 2026x–y).

The Factor Hierarchy Law exhaustively covers all six possible forms of factor hierarchies—from single-layer systems to cyclic regulation systems, from pure regime factors to semi-continuous regime factors (Tang, 2026g, §6). Each form has been empirically verified across six disciplines.

1.3 From the Factor Hierarchy Law to Information Isomorphism Epistemology

The discovery of the Factor Hierarchy Law gave rise to two methodological consequences.

First, since some factors function indirectly (regulating the exposure coefficients of other factors through interaction effects), how can one reliably distinguish regime factors from execution factors? How can one confirm that a detected regime switch is real rather than a statistical artifact? The Norms of Testability (Tang, 2026g, Chapter 3; Tang, 2026z) constitute the systematic answer to this question. They contain six functional dimensions—search completeness, regulatory relationships, structural breakpoints, causal direction, problem completeness, and process closure—forming a set of standardized testing procedures for regime switches. Functional dimensions remain relatively stable while implementation tools are replaceable.

Second, it was necessary to establish a principle to guarantee the reliability of any claim: the Principle of Testability (Tang, 2026g, Chapter 2)—any claim about factor importance in multi-factor systems must pass

complete regime-dependence tests before being elevated from hypothesis to reliable knowledge. Claiming that "this factor is important" without specifying "under which regime" is an incomplete proposition.

Through repeated deployment across six disciplines, an even more fundamental law was interrogated from the success of the operating system. If statistical structures did not contain information traces of regime switches, the six functional dimensions could not possibly have succeeded repeatedly across six completely independent disciplines. Therefore, the very success of the operating system proves a more fundamental ontological fact: statistical structures carry information about regime structures.

This is the Law of Information Isomorphism (Tang, 2026ac). Its argumentative path is not induction from finite instances to universal principles, but logical interrogation from the cross-disciplinary success of the scientific discovery operating system to its necessary premises. On the 2D Ising exact solution, two information-theoretic dualities provided precision verification—the Chow F achieves a globally unique maximum at the precise T_c , with deviation strictly zero in the analytic sense; the intercept/slope ratio is strictly monotonic with distance from T_c ($\rho=1.0000$). On the 3D Ising model, the algorithm automatically discovered a previously unknown double-peak fission structure.

The Law of Information Isomorphism is a dual meta-law: it is simultaneously an ontological law (statistical structures carry information about regime structures) and an epistemological meta-law (it answers how we can know the world). It provides the epistemological foundation for the Factor Hierarchy Law—it reveals "why the Factor Hierarchy Law could be discovered." The two are complementary: the Factor Hierarchy Law reveals how the world is organized, and the Law of Information Isomorphism reveals how we can know the world.

Thus, Information Isomorphism Epistemology was established as a complete epistemological system. Its four components—the Law of Information Isomorphism, the Factor Hierarchy Law, the Principle of Testability, and the Norms of Testability—each independently originated, together constituting a complete system from ontology to epistemological standards to methodology.

1.4 The Philosophical Consequences After the System's Completion

This system was not initially aimed at refuting agnosticism. Its starting point was a concrete, technical methodological question. But after its completion, a previously unforeseen philosophical consequence emerged: within the domain of scientific practice, the Law of Information Isomorphism replaces the function of the "thing-in-itself" as a specific explanatory item; the Principle of Testability blocks the retreat of "this is merely your subjective illusion"; the Norms of Testability provide a complete set of testing procedures and negative controls, making scientific practice equipped with this operating system no longer need to appeal to an unknowable "thing-in-itself" to guarantee its objectivity. The self-sufficiency of the phenomenal realm is already sufficient to support all the needs of scientific practice.

More importantly, this system has repeatedly demonstrated across six disciplines that it possesses a capability that agnosticism can never provide: the capability to actively discover unknown structures. The core logic of agnosticism—"whatever you discover, it is still merely phenomenon"—renders it incapable of actively discovering any new structure. Information Isomorphism Epistemology, in every discipline it deployed into, actively discovered new structures that traditional theories could not accommodate or explain. This discovery capability is not an accessory to philosophical argumentation—it is the fundamental reason why Information Isomorphism Epistemology is more powerful than agnosticism in scientific practice.

The task of this paper is to systematically demonstrate how this philosophical consequence naturally emerges from the establishment process of Information Isomorphism Epistemology. This paper does not attempt to logically falsify Kant's transcendental agnosticism—because transcendental arguments are not directly threatened by empirical evidence. The strategy of this paper is to demonstrate that when scientific practice is equipped with the operating system of Information Isomorphism Epistemology, the "thing-in-itself" is no longer necessary.

2. Information Isomorphism Epistemology: The Complete Division of Labor Among Four Components

Information Isomorphism Epistemology consists of four components. Each has an independent origin, and each bears an irreplaceable demonstrative role in refuting agnosticism.

2.1 The Law of Information Isomorphism

The Law of Information Isomorphism (Tang, 2026ac) reveals that in complex systems based on objective data and exhibiting regime switches, statistical structures carry information about regime structures. Its argumentative path is not induction but logical interrogation from the cross-disciplinary success of the operating system to its necessary premises. In refuting agnosticism, the Law of Information Isomorphism directly replaces the function of the "thing-in-itself" as a specific explanatory item—"Why are the laws of nature exactly thus? Because the structure of the thing-in-itself is thus" is replaced by "Because statistical structures carry information about regime structures." The Law of Information Isomorphism is a dual meta-law: it is simultaneously an ontological law and an epistemological meta-law. It provides the epistemological foundation for the Factor Hierarchy Law, and the two are complementary.

2.2 The Factor Hierarchy Law

The Factor Hierarchy Law (Tang, 2026g, 2026r) reveals that any complex system is organized by a hierarchical structure of regime factors and execution factors. It emerged from financial data and was repeatedly

verified across five completely independent disciplines. It exhaustively covers all six possible forms of factor hierarchies, each empirically verified across six disciplines. The Factor Hierarchy Law is a meta-law of meta-laws—it subsumes derived meta-laws such as Noether's theorem, the principle of least action, and Shannon's information theory, providing them with a unified structural template. At the same time, it is itself also a meta-law—revealing that any complex system is organized by a hierarchical structure of regime factors and execution factors. In refuting agnosticism, the Factor Hierarchy Law provides the ontological foundation—it reveals the specific content of the information being carried.

2.3 The Principle of Testability

The Principle of Testability (Tang, 2026g, Chapter 2) stipulates that any claim about factor importance in multi-factor systems must pass complete regime-dependence tests before being elevated from hypothesis to reliable knowledge. In refuting agnosticism, the Principle of Testability blocks the retreat of "this is merely your subjective illusion"—both the Law of Information Isomorphism and the Factor Hierarchy Law have already passed regime-dependence tests.

2.4 The Norms of Testability

The Norms of Testability (Tang, 2026g, Chapter 3; Tang, 2026z) contain six functional dimensions: search completeness, regulatory relationships, structural breakpoints, causal direction, problem completeness, and process closure. In refuting agnosticism, the Norms of Testability provide complete testing procedures and negative controls, excluding all alternative explanations.

2.5 The Complete Division of Labor Among the Four Components

Each of the four components independently bears a demonstrative dimension, and together they form a complete refutation system. The Law of Information Isomorphism replaces the explanatory function of the "thing-in-itself." The Factor Hierarchy Law reveals the specific content of the information being carried. The Principle of Testability blocks the retreat of "subjective illusion." The Norms of Testability exclude all alternative explanations. Every retreat of agnosticism falls within the adjudicative scope of Information Isomorphism Epistemology.

3. The Originality of Information Isomorphism Epistemology and Its Philosophical Consequences

3.1 The Overall Originality of Information Isomorphism Epistemology

Each component of Information Isomorphism Epistemology is original. The Factor Hierarchy Law is an original discovery—it was not derived from philosophical speculation but emerged from financial data. The Law of Information Isomorphism is an original discovery—it was interrogated as a necessary premise from the cross-disciplinary success of the operating system. The Principle of Testability is an original construction—not because the concept of "testing" is new, but because the standardized requirement that "a claim must specify under which regime" is new. The Norms of Testability are an original construction—the systematic integration of six functional dimensions is new. The four components together constitute a complete original system.

3.2 Discovery Capability: Post-Hoc Verification and Ex-Ante Discovery

Information Isomorphism Epistemology is not merely an epistemological theory—it is a scientific discovery operating system that has repeatedly demonstrated its discovery capability across six disciplines. These discoveries fall into two categories, with different philosophical force against agnosticism.

The first category consists of post-hoc verifications—detecting precise statistical signals on truth-known systems. In particle physics, locking onto the positions of known particles with professional phenomenological fitting precision; on the 2D Ising exact solution, the Chow F achieving a globally unique maximum at the precise T_c with deviation strictly zero. These discoveries prove the precision of the Law of Information Isomorphism.

The second category consists of ex-ante discoveries—actively discovering previously unknown real structures on truth-unknown systems. The 3D Ising double-peak fission is an ex-ante discovery: the algorithm, without any physical knowledge, actively discovered a hidden structure never previously revealed by traditional analysis—the Bai-Perron double breakpoints precisely aligned with the switch of spontaneous magnetization M_{sp} and the switch of the Binder cumulant U_4 . The Tang Break is an ex-ante discovery: stars exhibit a universal regime boundary at 4762K, with five independent physical dimensions all structurally ruptured, directly challenging the standard planet radius assumption. The chemical predictions are ex-ante discoveries: the operating system provided purely deductive numerical predictions for yet-unsynthesized elements, with all parameters locked from preceding elements. The $N=14$ statistical rupture is an ex-ante discovery: a position never previously marked by traditional nuclear physics was cross-validated by three independent tools as a statistically significant structural rupture.

Ex-ante discoveries constitute the most direct strike against agnosticism. What agnosticism claims is precisely that "objective data do not contain reliable information about deep organizational patterns." If this

claim were true, an algorithm could not possibly, without any physical knowledge, actively discover a previously unknown real structure that was later independently verified. Agnosticism can easily respond to post-hoc verification: "this is still merely phenomenon, you have simply discovered more precise regularities within the phenomenal realm." But it struggles to respond to ex-ante discovery—because ex-ante discovery is not passively verifying known structures but actively penetrating unknown territory.

3.3 Tool Replacement, Not Functional Stripping

In scientific practice equipped with Information Isomorphism Epistemology, the "thing-in-itself" no longer bears any indispensable explanatory function. This is not because the function of the "thing-in-itself" was "discovered" to be superfluous—but because better epistemological tools have been developed, and it is no longer needed. Information Isomorphism Epistemology not only provides more precise, more testable tools, but more importantly, it provides something that the concept of the "thing-in-itself" can never provide: the capability to actively discover unknown structures. After the upgrade of epistemological tools, the old conceptual framework is not defeated but replaced.

4. The Tool Replacement of the "Thing-in-Itself" Within the Operating System

4.1 The Two Levels of Functions of the "Thing-in-Itself"

The "thing-in-itself" in Kant's system bears two different levels of functions. The first level: the necessary fulcrum of transcendental logic—Kant's sensibility is passively receptive; something must be given for phenomena not to be illusions. The second level: an explanatory item for concrete scientific discoveries—why are the laws of nature exactly thus? "Because the structure of the thing-in-itself is thus."

4.2 The Direct Replacement of the Second-Level Function by the Law of Information Isomorphism

For the "thing-in-itself" as a specific explanatory item, the Law of Information Isomorphism constitutes a direct replacement. "Why are the laws of nature exactly thus? Because the structure of the thing-in-itself is thus." The Law of Information Isomorphism answers: "Because statistical structures carry information about regime structures. This answer requires no presupposition of any unknowable external source—it depends only on testable relations within the phenomenal realm." The replacement by the Law of Information Isomorphism at this level is complete.

4.3 The Tool Replacement of the First-Level Function by the Principle of Testability and the Norms of Testability

For the "thing-in-itself" as the fulcrum of transcendental logic, this paper candidly acknowledges that Information Isomorphism Epistemology cannot and does not attempt to replace this function. However, the Principle of Testability and the Norms of Testability together render this question no longer in need of being answered within scientific practice. The foundation of scientific knowledge is not "phenomena faithfully reflect the thing-in-itself"—but "information about regime structures can be precisely recovered within the operating system." Furthermore, this paper can turn the question back upon philosophers who insist on the concept of the "thing-in-itself": please provide an operationalized definition of "the structure of the thing-in-itself" such that it differs from "regime structure," and such that this difference has any testable consequences in scientific practice. If the other party cannot provide such a definition, then the concept of "the structure of the thing-in-itself" is empty in scientific practice.

4.4 Boundary Dissolution, Not Logical Refutation

A Kantian could point out that one of Kant's intentions in introducing the "thing-in-itself" was precisely to demarcate the legitimate domain of scientific knowledge. This is a legitimate philosophical response. This paper does not attempt to logically falsify it. The strategy of this paper is pragmatic dissolution: once it is established that "the self-sufficiency of the phenomenal realm is already sufficient to support all the needs of scientific practice," the urgency of the question "what lies beyond the boundary" is dissolved—not that it becomes a pseudo-problem, but that it is downgraded from "must be answered" to "can be set aside." The core contribution of Information Isomorphism Epistemology is precisely to push this boundary one step inward in the sense of scientific practice—transforming the "thing-in-itself" from a "boundary concept that must be presupposed" into a "metaphysical preference that can be set aside."

4.5 The Principle of Testability as a New Boundary

Kant drew a line using transcendental argumentation—beyond that line lies the "unknowable thing-in-itself." This line does not require Kant himself to pass any test, because it is transcendental. Information Isomorphism Epistemology also draws a line—between "claims that can be tested by regime dependence" and "claims that cannot be tested by regime dependence." Any proposition that claims facts about the world must pass regime-dependence tests. This new line treats all claims equally. The Law of Information Isomorphism has already passed the test. Agnosticism claims that "the thing-in-itself is forever unknowable"—this is a claim about facts in the world, but it cannot pass regime-dependence tests. Therefore, it has no legitimacy in scientific practice. This is not refuting Kant within the boundary he demarcated—this is redrawing a new boundary. The

reason this new line is more powerful than Kant's line is not that it is sharper, but that it is fairer. It requires all claims—including its own—to submit to the same tests.

4.6 The Epistemological Status of the Principle of Testability

Is the Principle of Testability itself an untestable metaphysical presupposition? The response of this paper is: the Principle of Testability is not an externally imposed metaphysical presupposition but an operational necessity that emerged internally from the discovery of the Factor Hierarchy Law. When Tang (2026e) discovered that "the function of some factors is not to directly explain the dependent variable but to regulate the exposure coefficients of other factors," a methodological question naturally arose: how to distinguish these two types of factors? The Principle of Testability is the answer to this question. It is not "I decree that science must do this" but "if I do not do this, even I myself cannot distinguish true discoveries from false positives." More importantly, the Principle of Testability bears the risk of being empirically overturned—if a system is discovered in the future in which regime switches can be reliably confirmed without any testing, the scope of applicability of this principle would be revised. It is not unfalsifiable.

4.7 The Realist Status of Regime Factors and Execution Factors

A structural realist might ask: if regime switches themselves are real, is the distinction between "regime factors" and "execution factors" real, or merely a product of the model? The response of this paper is: on the 2D Ising exact solution, the dichotomy is analytically necessary, not a matter of model choice. On either side of the precise T_c , the intercept and slope of the $M \sim h$ equation undergo systematic changes—this is not a choice of model fitting but an inevitable consequence of the precise mathematical structure. Temperature as a regime factor, external magnetic field as an execution factor—this dichotomy is a property of the physical system itself. On truth-unknown systems, the dichotomy is a realist hypothesis that bears the risk of falsification. If in a new discipline, all candidate factors show no significant interaction effects while all important factors manifest as direct effects, the applicability of the Factor Hierarchy Law in that discipline would be weakened.

4.8 The Realist Anchor: Responding to Global Constructive Skepticism

A skeptic could pose an even more fundamental challenge: "Your testing procedures are themselves an elaborate illusion-generating system." This paper acknowledges the philosophical legitimacy of this skepticism and provides the following response. On the 2D Ising exact solution, the "regime structure" detected by the operating system is not an illusion—because the exact solution was derived algebraically by Onsager in 1944, long before the invention of any structural breakpoint detection tools. A more important realist anchor is the active discovery of the 3D Ising double-peak fission: the algorithm, without any physical knowledge, actively discovered a previously unknown real physical structure—the Bai-Perron double breakpoints precisely aligned

with the switch of spontaneous magnetization M_{sp} and the switch of the Binder cumulant U_4 . If the operating system merely produced illusions, it could not have actively discovered a real physical structure on a truth-unknown system that was later independently verified. This anchor does not provide absolute metaphysical guarantees—but it provides the strongest epistemological guarantee in scientific practice: it holds strictly in systems that can be independently verified.

4.9 Information Isomorphism Epistemology and Transcendental Philosophy

A transcendental philosopher could argue that the very possibility of the Norms of Testability presupposes certain conditions. The response of this paper is: Information Isomorphism Epistemology does require certain conditions to operate—data must be objective, and systems must exhibit regime switches—but these conditions are not "transcendental" but empirical, and can themselves be empirically tested. More importantly, the division of labor between Information Isomorphism Epistemology and transcendental philosophy is being redrawn: not "transcendental philosophy provides legitimacy, and empirical science operates within the bounds of that legitimacy"—but "empirical science discovers regularities in its own practice, and then uses those regularities to redraw the boundary between science and metaphysics." If transcendental philosophers insist that this very framework also requires transcendental conditions to guarantee it, the response of this paper is: please provide criteria by which these transcendental conditions have any testable consequences in scientific practice. If they cannot provide such criteria, then these transcendental conditions are empty in scientific practice—they can be set aside.

5. The Reconstruction of Knowability: An Open, Evolvable, Falsifiable Knowability

5.1 The Knowability of Information Isomorphism Epistemology Is Not Dogmatism

Traditional knowability—from Descartes to Hegel—claimed that the world itself can be directly grasped by reason. This is a form of dogmatism: it asserts that human cognitive capacity has no boundaries. The knowability reconstructed by Information Isomorphism Epistemology does not claim that human cognitive capacity is unlimited, nor that we have touched ultimate reality. It claims only that in objective data systems satisfying the relevant conditions, information about regime structures can be precisely recovered from statistical structures. This claim has been rigorously proven on truth-known systems and independently tested on truth-unknown systems.

5.2 Open, Risk-Bearing, Discovery-Capable Knowability vs. Closed, Risk-Immune, Discovery-Incapable Agnosticism

Kant's agnosticism is closed. "The thing-in-itself is forever unknowable"—this is a terminal assertion that will never be altered by any future discovery. The knowability of Information Isomorphism Epistemology is open—functional dimensions may expand or contract, and questions that cannot be answered today may be answerable after future upgrades of the operating system. More critically, Information Isomorphism Epistemology explicitly provides falsification conditions and bears the risk of being empirically overturned, while Kant's agnosticism bears no risk of being overturned.

But the most fatal contrast lies in discovery capability. Agnosticism cannot actively discover any unknown structure—because its core logic is precisely "whatever you discover, it is still merely phenomenon." Information Isomorphism Epistemology, in every discipline it deployed into, actively discovered new structures that traditional theories could not accommodate. Particularly the ex-ante discoveries on truth-unknown systems—the 3D Ising double-peak fission, the Tang Break, the chemical predictions, the $N=14$ statistical rupture—each was a real physical structure never previously recognized by humanity before the deployment of the operating system. This discovery capability is not an accessory to philosophical argumentation; it is the fundamental reason why Information Isomorphism Epistemology is more powerful than agnosticism in scientific practice.

5.3 The Coordinates of Information Isomorphism Epistemology's Knowability in Contemporary Philosophy

The knowability of Information Isomorphism Epistemology can be positioned as a form of "operational structural realism." It shares with structural realism the core intuition that "structure is reality," but expands "structure" from static mathematical equations to dynamic regime-switching relations—structure is not an unchanging equation form but a dynamic relational network organized by the hierarchy of regime factors and execution factors, undergoing regime switches at critical values. It shares with pragmatist knowability the spirit of open evolution and self-correction, but replaces the "long-run convergence" promise with a set of standardized testing procedures. It shares with naturalized epistemology the approach of embedding epistemology within empirical science, but retains the Principle of Testability as an irreducible normative anchor.

5.4 The Reconstruction of Knowability

Information Isomorphism Epistemology lays the foundation for a new type of knowability. This knowability does not claim to answer every "why," nor to have touched ultimate reality. But it confirms that before every concrete objective data system, human reason has the capacity to penetrate the chaos of

phenomena and discover the objective organizational structures within the phenomenal realm. This capacity requires no metaphysical guarantee—it requires only the rigorous adherence to the four components of Information Isomorphism Epistemology.

6. Candid Declarations

The entire empirical foundation of this paper comes from a series of preprints by the same author (Tang, 2026a–2026ab) and has not yet undergone independent third-party replication. The self-sufficiency of the phenomenal realm revealed by Information Isomorphism Epistemology is currently based on convergent evidence from six disciplines. Broader cross-disciplinary verification requires the participation of independent researchers.

This paper does not claim to have logically falsified Kant's transcendental idealist system. This paper does not claim to have replaced the function of the "thing-in-itself" as the fulcrum of transcendental logic. The strategy of this paper is to demonstrate that in scientific practice equipped with Information Isomorphism Epistemology, the "thing-in-itself" no longer bears any indispensable explanatory function. Some individual dimensions within the Norms of Testability have precedents in statistics and econometrics. The original contribution of this paper lies in systematically integrating these dimensions into a standardized, mandatory, complete testing procedure oriented toward regime switch detection, and in proving its cross-disciplinary universality.

The "regime structure" of Information Isomorphism Epistemology is an operational definition—it is precisely that which is revealed by the structural breakpoints, interaction effects, and regime-execution hierarchies detected through the six functional dimensions. The knowability of Information Isomorphism Epistemology does not claim to transcend phenomena—it claims only that within the phenomenal realm, there exist objective organizational structures that can be precisely decoded. This paper does not claim that this operational definition is equivalent to Kant's thing-in-itself.

The Factor Hierarchy Law is a meta-law of meta-laws, and is simultaneously itself a meta-law—the former is its original positioning in Paper 18, and the latter describes its content of revealing that "any complex system is organized by a hierarchical structure of regime factors and execution factors." The Law of Information Isomorphism is a dual meta-law, providing the epistemological foundation for the Factor Hierarchy Law; the two are complementary. This paper's understanding of Kantian agnosticism is based on standard interpretations of the *Critique of Pure Reason* and the *Critique of the Power of Judgment*. The systematic dialogue between Information Isomorphism Epistemology and contemporary philosophical traditions such as structural realism, pragmatism, and naturalized epistemology is a core direction for subsequent work.

7. Conclusion

This paper has presented the establishment process of an original epistemological system—Information Isomorphism Epistemology—and its philosophical consequences.

The establishment of this system was not aimed at refuting agnosticism. Its starting point was a concrete methodological question. But from this concrete question, the Factor Hierarchy Law first emerged from financial data. This discovery gave rise to the Principle of Testability and the Norms of Testability. Through repeated deployment across six disciplines, the Law of Information Isomorphism was interrogated from the success of the operating system. Thus, Information Isomorphism Epistemology was established as a complete epistemological system.

After the completion of this system, a previously unforeseen philosophical consequence emerged. The core argument of this paper can be summarized as "tool replacement"—not proving Kant wrong through philosophical debate, but using an original epistemological operating system to render the "thing-in-itself" no longer bearing any indispensable function in scientific practice. The Law of Information Isomorphism replaces the function of the "thing-in-itself" as a specific explanatory item. The Factor Hierarchy Law reveals the universal organizational patterns within the phenomenal realm. The Principle of Testability blocks the retreat of "this is merely your subjective illusion" while simultaneously drawing a new boundary that treats all claims equally. The Norms of Testability provide a complete set of testing procedures and negative controls.

More importantly, Information Isomorphism Epistemology has repeatedly demonstrated across six disciplines that it possesses a capability that agnosticism can never provide: the capability to actively discover unknown structures. Particularly the ex-ante discoveries on truth-unknown systems—the 3D Ising double-peak fission, the Tang Break, the chemical predictions, the $N=14$ statistical rupture—each was never previously recognized by humanity before the deployment of the operating system. The core logic of agnosticism renders it incapable of actively discovering any new structure. Information Isomorphism Epistemology, in every discipline it deployed into, actively discovered new structures that traditional theories could not accommodate or explain. This discovery capability is not an accessory to philosophical argumentation—it is the fundamental reason why Information Isomorphism Epistemology is more powerful than agnosticism in scientific practice.

Information Isomorphism Epistemology thus lays the foundation for a new type of knowability. This knowability is not dogmatism—it does not claim to have already answered everything. It is open, evolvable, risk-bearing, and discovery-capable. In scientific practice equipped with Information Isomorphism Epistemology, knowability no longer requires any metaphysical guarantee. It requires only the rigorous adherence to the four components of Information Isomorphism Epistemology. This is not philosophical speculation defeating philosophical presupposition. This is the upgrade of epistemological tools rendering old metaphysical presuppositions unnecessary.

Data Availability Statement

This paper is a theoretical overview and philosophical argumentation paper. All empirical data come from the series of preprints by Tang (2026a–2026ab). The two information-theoretic dualities on the 2D Ising exact solution are detailed in Tang (2026k) and Tang (2026x) Theorem A. The discovery process of the Factor Hierarchy Law is detailed in Tang (2026e) and Tang (2026g). All data are completely public, and any researcher can independently replicate all results.

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